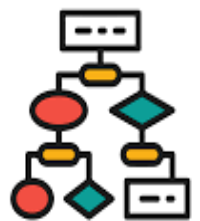


Project

UNSUPERVISED LEARNING



Content:

1. EDA

1.1 Data Context.

1.2 Univariate analysis and Bivariate analysis.

1.3 Visualizations to identify the patterns.

1.4 Observations and Relationship between variables.

2. Data Preprocessing

2.1 Missing Value Treatment.

2.2 Outlier Detection and Treatment.

2.3 Feature Engineering.

2.4 Data Scaling.

3. K-means Clustering

3.1 Apply K-means Clustering.

3.2 Plot the Elbow curve.

3.3 Check Silhouette Scores.

3.4 Figure out the appropriate number of clusters.

3.5 Cluster Profiling.

4. Hierarchical Clustering

4.1 Apply Hierarchical clustering with different linkage methods.

4.2 Plot dendrograms for each linkage method.

4.3 Check cophenetic correlation for each linkage method.

4.4 Figure out the appropriate number of clusters.

4.5 Cluster Profiling.

5. K-means vs Hierarchical Clustering

Data Context:

The dataset of customers with attributes including credit limits, number of cards, and frequencies of contact via bank visits, online logins, and phone calls.

Objective: Analyze the spending patterns and interaction behavior of AllLife Bank's credit card customers to identify distinct customer segments.

Enable the bank to personalize marketing campaigns for better market penetration and effective upselling, and to optimize service delivery channels (branch visits, online, and call support) for faster query resolution and improved customer experience.

EDA and Data Preprocessing

Observation

- The dataset contains 660 records.
- All columns have an integer data type.
- There are zero missing values across all features.
- There are **655 unique keys** in a **660-row dataset**.

Insight

- The same Customer Key, one row shows high Total_visits_bank while the other shows high Total_visits_online.
- This suggests the customer's activity is being split across two database records rather than being unified.
- One record is likely old while the other is fresh.
- Found 5 duplicate keys in a dataset of 660, it confirms a 0.75% error rate in the data collection process.

Observation:

Metric	Avg Credit Limit	Online Visits	Calls Made
Mean	34,574	2.3	3.5
Maximum	200,000	15	10
Standard Deviation	High (34,654)	Moderate (2.7)	High (2.8)

Insight:

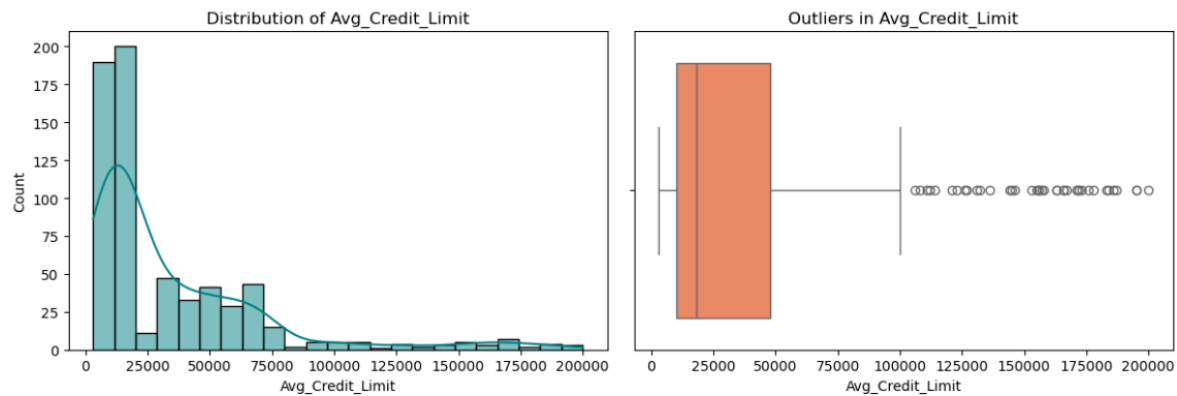
- Average Credit Limit: Difference between the Mean (34,500) and the Median (50%)18,000).
- The data is heavily right-skewed. A small group of high-value customers with very high credit limits up to 200,000 is pulling the average upward.
- Total Visits Bank: The average is low (approx 2.4), with a maximum of 5.
- Total Visits Online: This feature has the widest variance. Some customers never go online , while others visit 15 time.
- Total Calls Made: The max value of 10 suggests a "High Support" segment.
- The Spread: Most customers have between 4 and 6 cards.
- Insight: Having only 1 card or as many as 10 is an outlier behavior. Customers with 10 cards almost certainly correlate with the high-limit/online-heavy segment.

Observation:

Feature	Distribution (Histogram)	Outlier Presence (Boxplot)
Avg_Credit_Limit	Heavily right-skewed	High (many outliers)
Total_visits_bank	Fairly uniform / low variation	Low
Total_visits_online	Bi-modal (two peaks)	Moderate
Total_calls_made	Right-skewed	Moderate

Avg_Credit_Limit:

Insights:

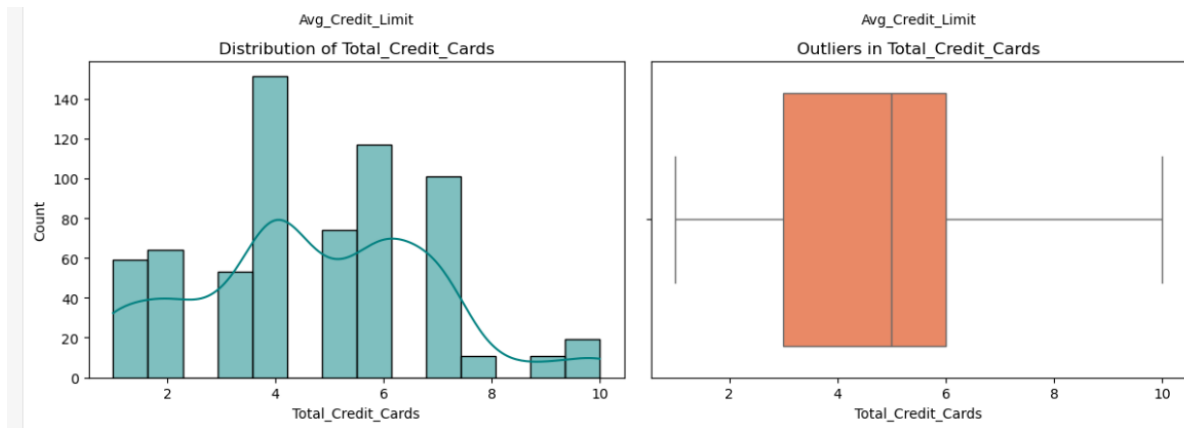


The histogram shows a heavy right skew. Most customers are clustered below 25,000, but a long "tail" stretches all the way to 200,000.

Outliers: The boxplot confirms a significant number of outliers beyond the 100,000 mark.

Business Insight: These outliers represent highest-value segment. In a 3-cluster model, these individuals will likely form their own "Elite" cluster because their financial profile is mathematically so distant from the "average" customer.

Total_Credit_Cards:

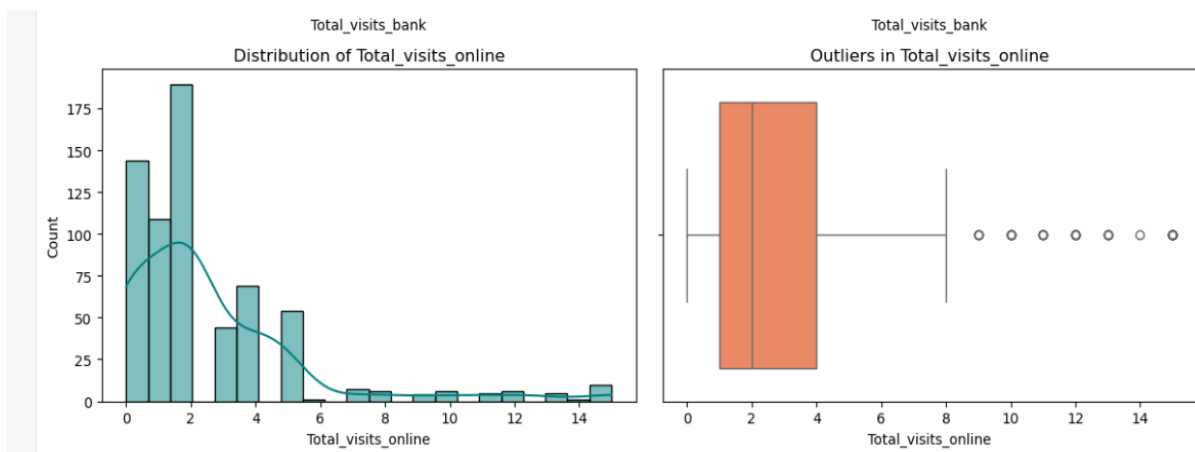


Multi-modal has multiple peaks. It has clearly see three distinct bumps around **2, 4–6, and 10** cards.

Outliers: Interestingly, the boxplot shows **zero outliers**. The data stays strictly within the 1–10 range.

Business Insight: This is a perfect setup for clustering. The three peaks suggest that customers naturally categorize themselves into "Low," "Medium," and "High" card users, which aligns perfectly with goal of finding three clusters.

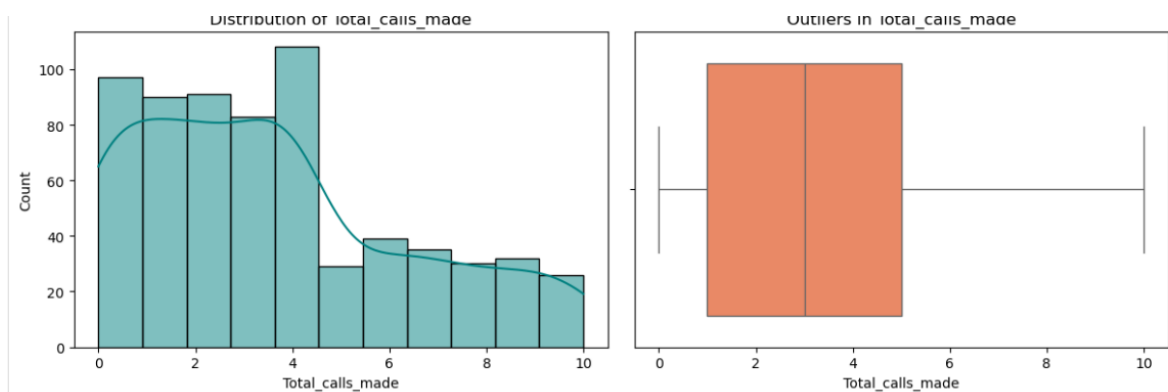
Total_visits_online:



Distribution: Shows a peak at 2 visits, with a fairly even spread across other values from 0 to 5.

Outliers: No outliers are detected. The narrow range suggests this feature identifies "Traditional" banking behavior without extreme deviations.

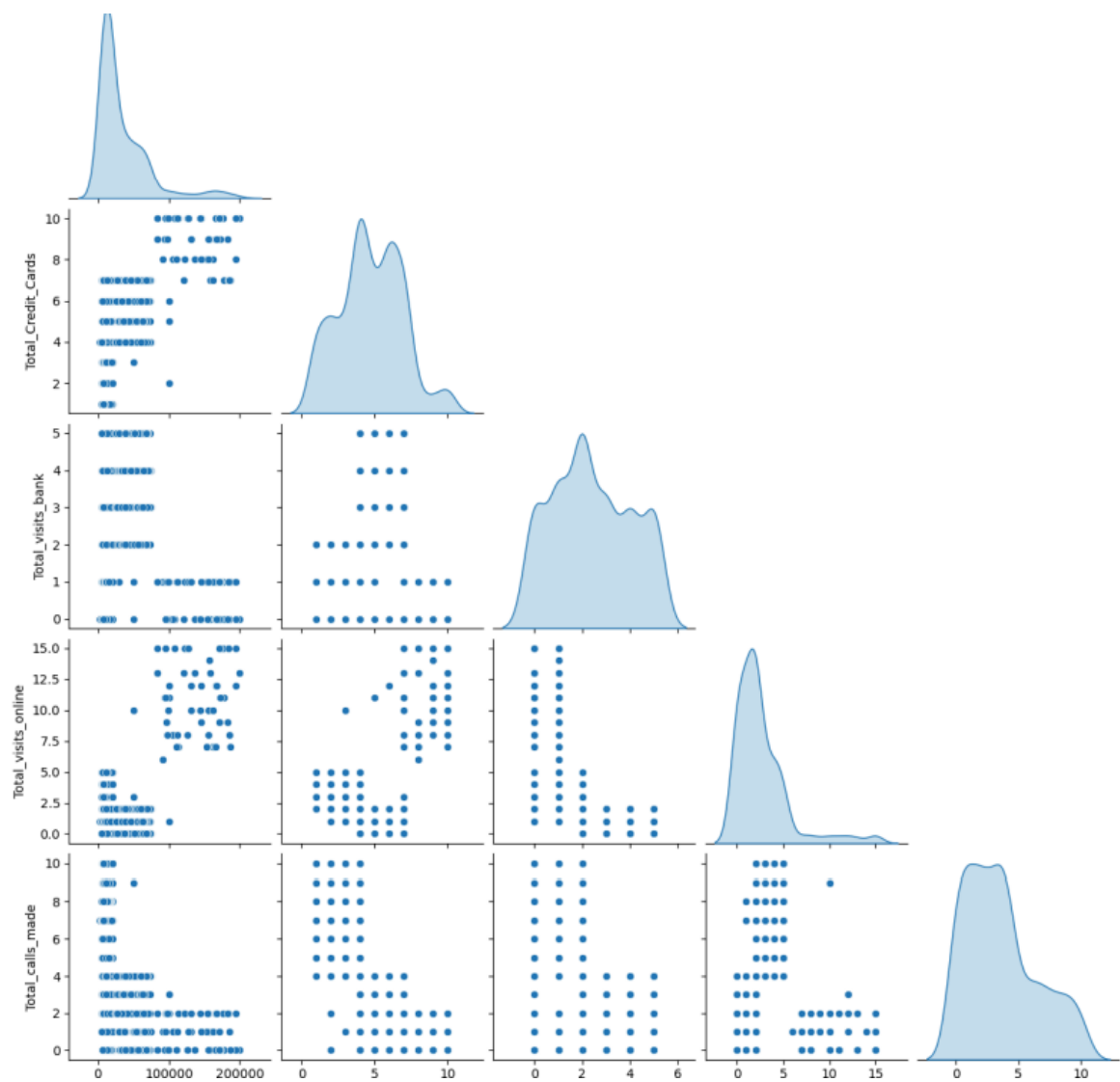
Total_calls_made



Distribution Shape: The histogram for Total_calls_made shows a distinct drop-off after 4 calls. While a large portion of the population makes 0–4 calls,

No Statistical Outliers: Interestingly, the boxplot shows no individual outliers. This means that making 10 calls.

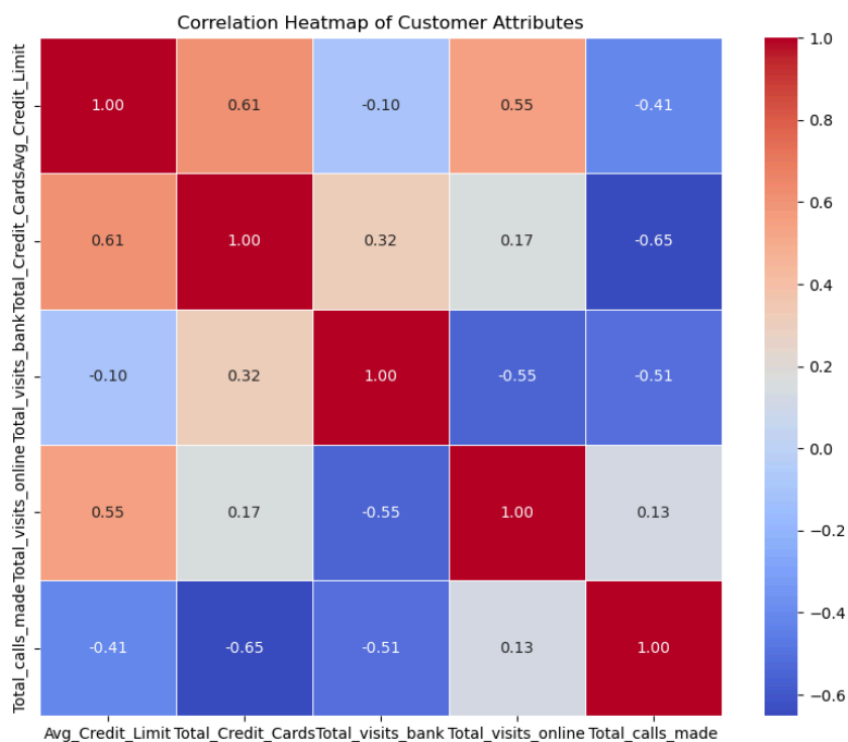
Operational Insight: High call volume is often either the customer finds the digital interface difficult or have complex issues that automated systems aren't solving.



Negative Correlations: The pairplot shows a clear inverse relationship between Total_visits_online and Total_calls_made.

Business Insight: Customers who are comfortable online rarely call the bank, and vice versa. This confirms that "Digital" vs. "Support" is a major dividing line for segmentation strategy.

The Branch Group: Total_visits_bank peaks at 2 visits and shows a fairly uniform distribution without outliers. This group appears middle-of-the-road in both credit limit and digital usage.



Data Insight: The Avg_Credit_Limit histogram shows a significant right skew, with a long tail of "outliers" extending up to 200,000. These high-limit customers correlate strongly with higher Total_visits_online (0.55 correlation) and more Total_Credit_Cards (0.61 correlation).

There is a strong negative correlation (-0.65) between Total_calls_made and Total_Credit_Cards. The pairplot and histogram show that customers who call the bank frequently (up to 10 times) typically have fewer cards and lower credit limits.

Total_visits_bank shows a peak around 2 visits and is negatively correlated with Total_visits_online (-0.55). These customers are likely more traditional in their banking habits and do not utilize digital channels as frequently.

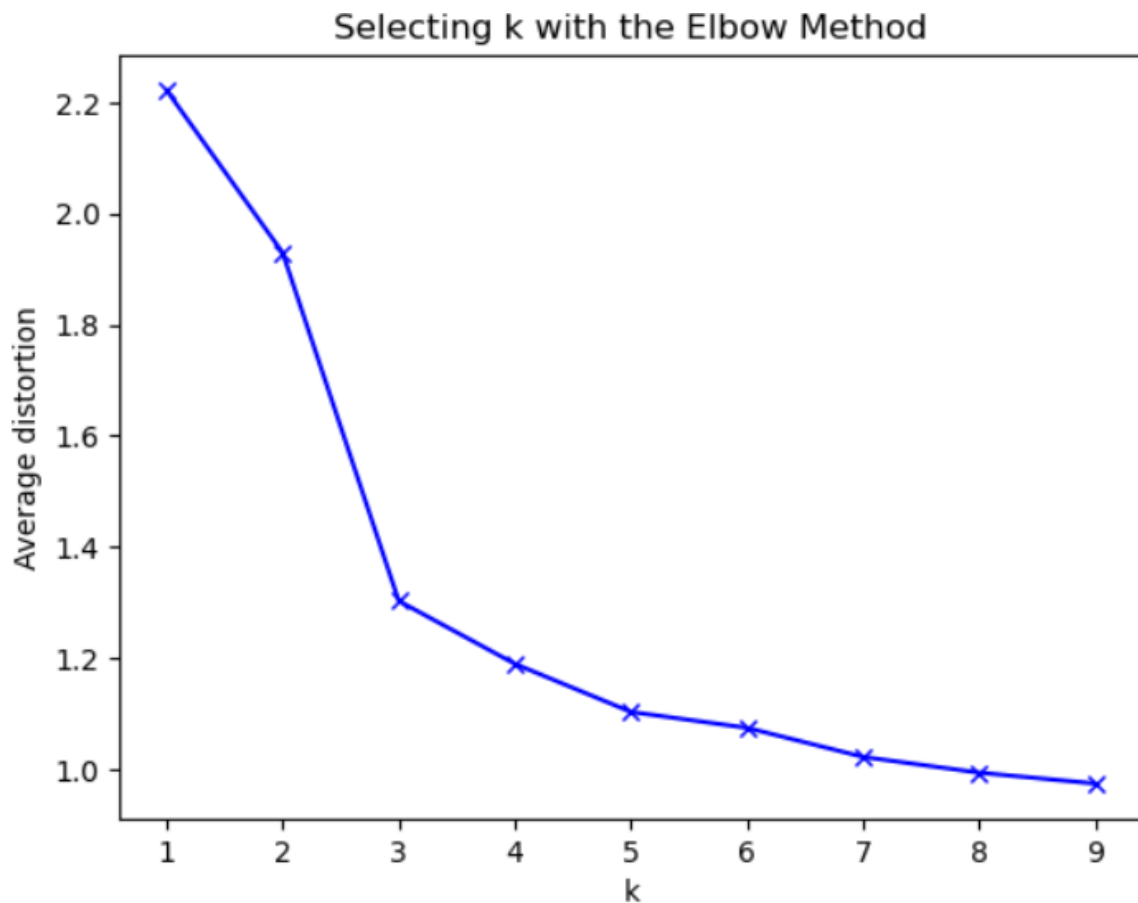
The Total_Credit_Cards distribution is multi-modal, with clear natural peaks at 4, 6, and 10 cards. Interestingly, this feature has zero outliers, indicating that customer behavior regarding product quantity is very stable and tiered.

Business Recommendation:

Tiered Product Bundles: Align marketing tiers (e.g., Silver, Gold, Platinum) with these natural peaks. A customer with 4 cards should be targeted for an "upgrade" to the 6-card tier through a curated bundle offer.

High-Touch Sales: Since these customers are physically present in branches, use those visits for high-trust product discussions, such as personal loans or insurance, which often require face-to-face interaction.

K-Means



Observation:

Segment	Primary Interaction	Financial Profile
One	High online usage (max 15)	High credit (up to 200k)
Second	Branch-focused (avg 2 visits)	Mid-tier credit
Three	High call volume (max 10)	Low credit / card users

The Bend at k=3: There is a very sharp drop in distortion moving from k=1 to k=3. After k=3, the line flattens out considerably.

This indicates that adding a 4th or 5th cluster doesn't provide enough additional "information" or grouping power to justify the extra complexity.

Data naturally organizes itself into three distinct populations. Any further subdivision would likely lead to "overfitting," where create segments that don't actually have unique behavioral differences.

Since the Elbow Method confirms $k=3$, should move forward with a 3-cluster strategy:

1. **Cluster 1 (Digital Elites):** High credit limits and high online engagement.
2. **Cluster 2 (Traditionalists):** Moderate credit and frequent branch visits.
3. **Cluster 3 (Support Seekers):** Low credit limits and high call volume.

The "Whale" Effect: The Avg_Credit_Limit distribution shows a heavy right skew, with a significant number of outliers extending up to 200,000.

High credit limits are strongly correlated with Total_Credit_Cards (0.61) and Total_visits_online (0.55).

Customers making up to 10 calls typically have fewer cards and lower credit limits, suggesting struggle more with the product or system.

The data clearly separates customers by how choose to interact with the bank:

Total_visits_bank is negatively correlated with Total_visits_online (-0.55), proving that base is split between "Traditionalists" and "Modernists".

Branch Utilization: Most customers visit the branch roughly 2 times, and there are no extreme outliers for physical visits.

The Elbow Method shows a sharp "bend" at $k=3$, indicating that customers naturally divide into three distinct behavioral archetypes.

Natural Card Peaks: The Total_Credit_Cards distribution shows three clear peaks around 4, 6, and 10 cards.

Recommendation:

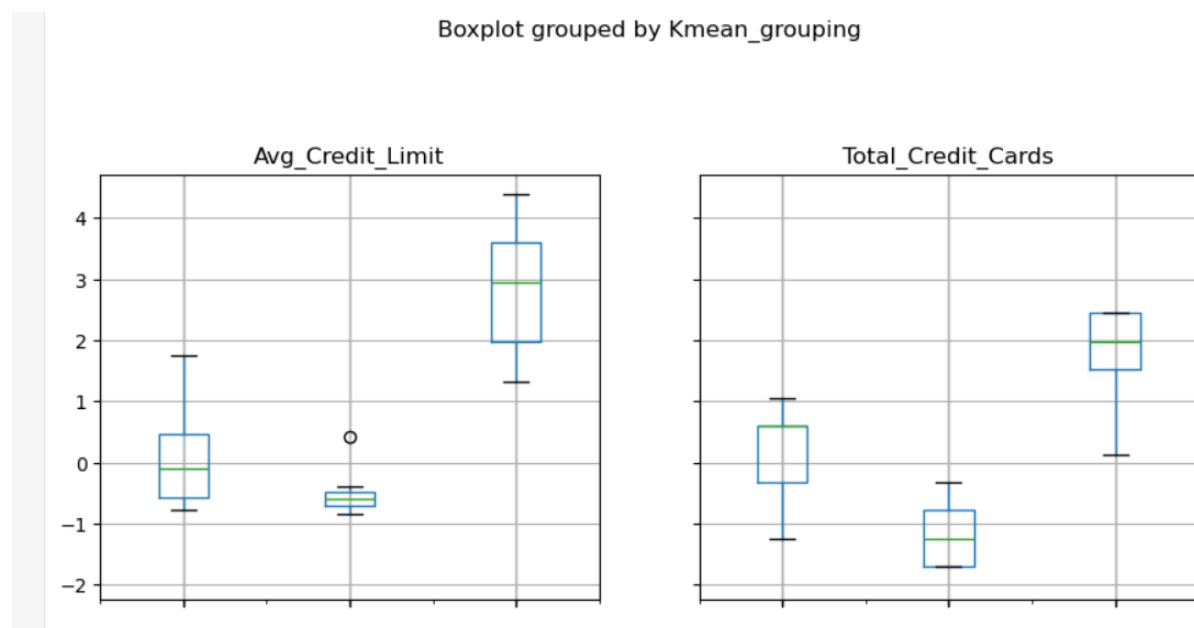
Put in exclusive digital-first rewards for this segment (e.g., luxury travel perks), as it is already high-frequency online users.

Offer high-limit card consolidation products to simplify their wallet while capturing a larger share of their total spend.

Digital Migration: Incentivize mobile deposits or online bill pay for the branch-reliant group to reduce high-overhead physical visits.

In-Person Cross-Selling: Use the predictable 2-visit window for high-trust product discussions (like mortgages) that benefit from face-to-face interaction.

Inverse Relationship: There is a strong negative correlation (-0.65) between Total_calls_made and Total_Credit_Cards.

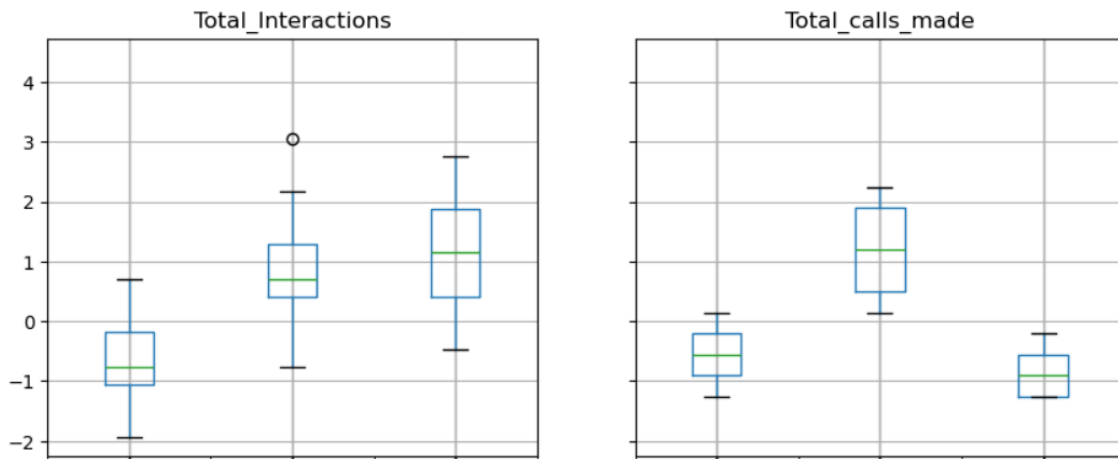


Observation :

Cluster (L → R)	Avg Credit Limit	Total Credit Cards
Cluster 0 (Left)	Average (~0)	Moderate
Cluster 1 (Center)	Low (negative)	Low
Cluster 2 (Right)	High (positive)	High

Insights :

Cluster 2 (Right) has much "taller" boxes. In "High Value" customers, there is more diversity in credit use compared to the very consistent behavior of the "Low Value" group in Cluster 1.



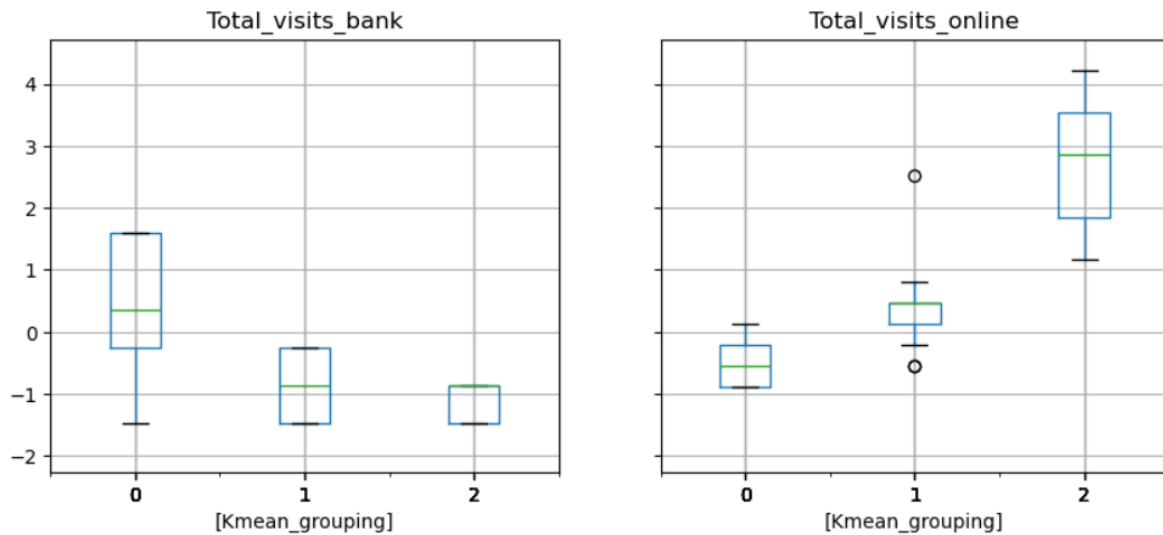
Observation:

Cluster	Interaction	Call made
Cluster 0 (Left)	Mid-range limits and cards	Low interactions / calls
Cluster 1 (Center)	Low limits and few cards	High calls and interactions
Cluster 2 (Right)	High limits and many cards	Moderate to high interactions

For Cluster 1: Investigate the reason for the high call volume. If it's for basic inquiries, could save costs by pushing them toward digital self-service tools.

For Cluster 0: This group is at risk of being "forgotten." Since they don't call, might need proactive marketing or loyalty offers to keep them from churning to a competitor.

For Cluster 2: Ensure their digital interaction points are seamless. Since they have high limits and high interactions, any friction in app or portal will impact most profitable segment the most.

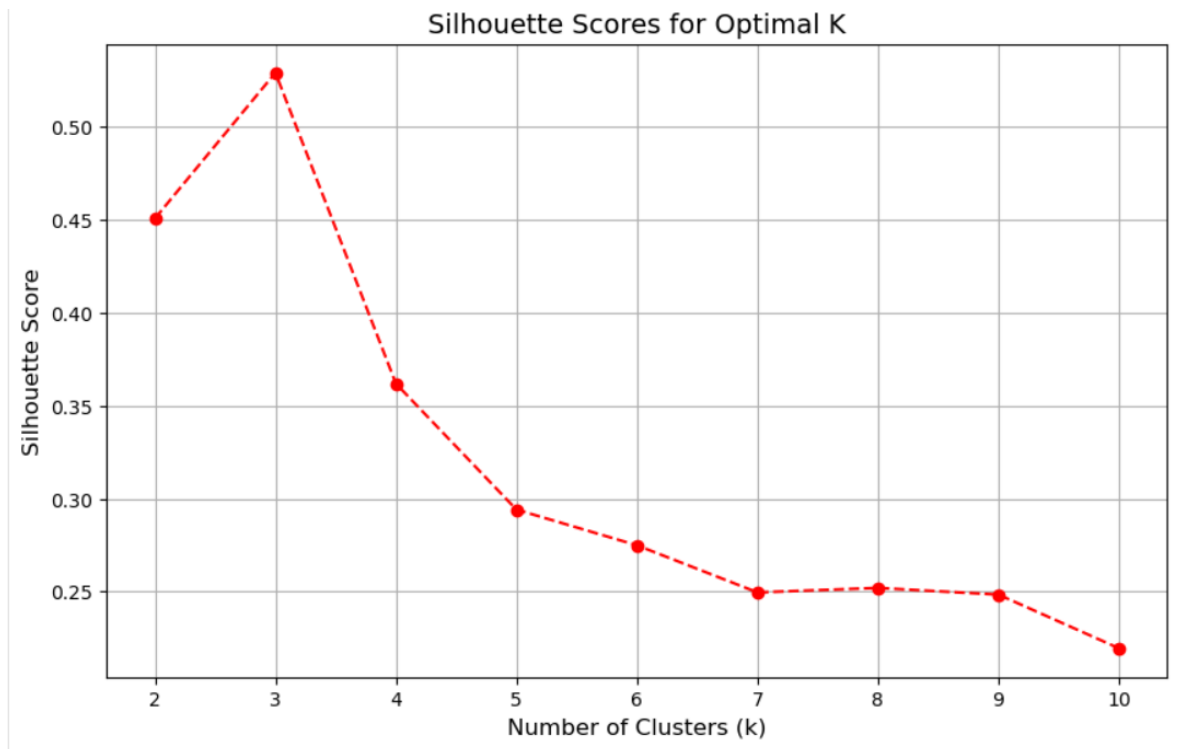


Cluster	Credit Strength	Primary Channel	Interaction Level
Cluster 0	Moderate	In-person (bank visits)	Low
Cluster 1	Low	Phone (calls)	High
Cluster 2	High	Digital (online visits)	High

High-limit customers don't want to step foot in a branch; want a flawless digital experience.

almost exclusively online, while Group 0 is the only segment with a significant physical presence in the bank.

low credit limits and high call volumes, they are likely "cost-center" customers. Calling for help or to resolve issues rather than visiting the bank or successfully using online tools.



> 0.7 : Indicates a "strong" cluster structure where points are well-separated and highly cohesive.

$0.5 < \leq 0.7$: Considered a "reasonable" or "good" clustering quality.

$0.25 < \leq 0.5$: Suggests a "weak" structure; clusters may be overlapping or poorly defined.

0 or Negative: Points near 0 lie on the boundary between clusters, while negative scores indicate likely misclassification.

High Score Cluster: These are clean (likely High-Value/Digital group). They are easy to target with marketing because behavior is very consistent.

Low Score Cluster: These Standard User group has a low silhouette score, it means they behave a bit like the other two groups, making them harder to define.

The Elbow Point: If $K=3$ isn't near the "elbow" of the inertia plot, over-simplifying or over-complicating the customer base.

Features where the boxes for different clusters do not overlap on the Y-axis (the scaled value) are primary differentiators.

A "short" box indicates high similarity within that group for that feature (low variance), while a "tall" box suggests a more diverse group.

Relative Positioning: Since the data is scaled, a value of 0 represents the dataset average. Values above 0 are "High" and below 0 are "Low" relative to the total population.

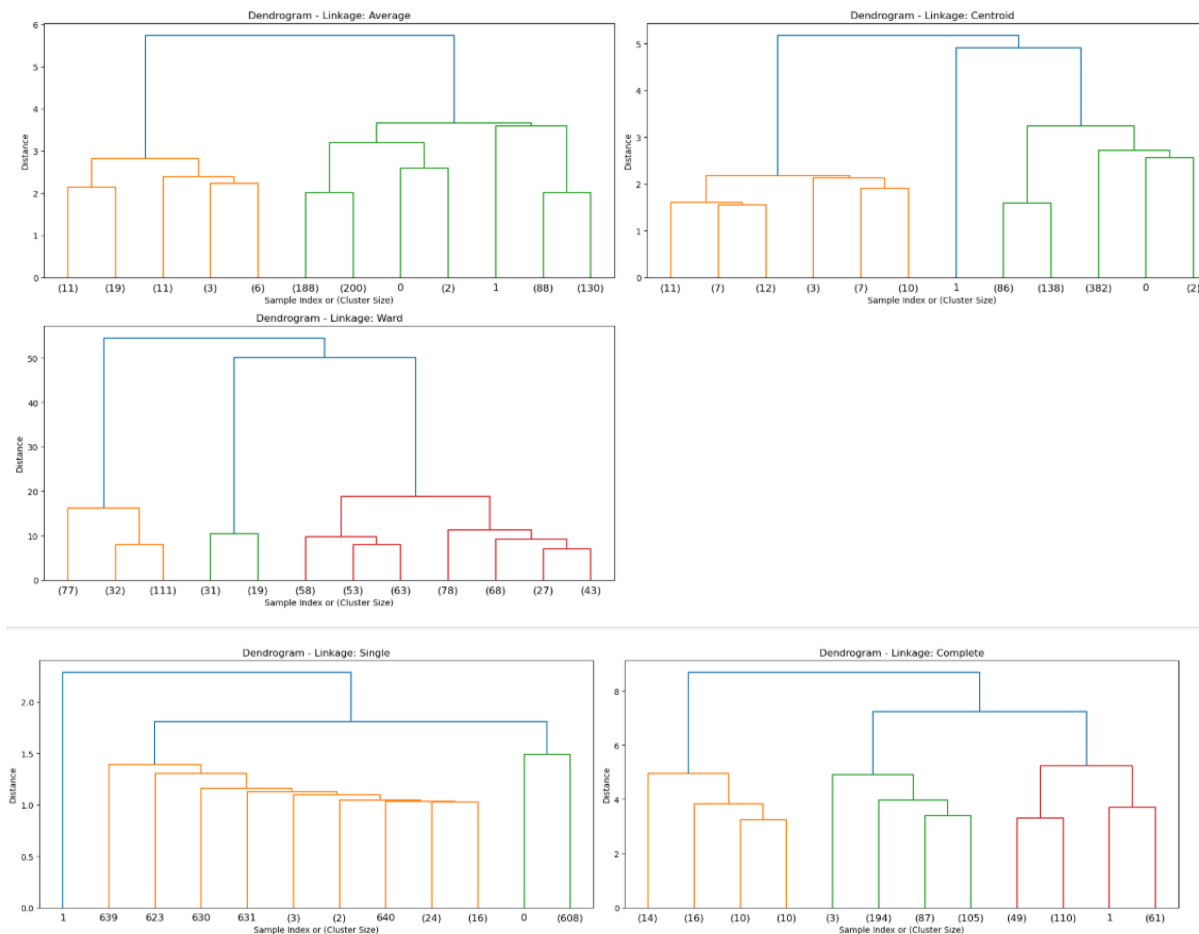
Metric	High-Value	Traditional	Support-Seeking
Avg_Credit_Limit	High (positive scaled value)	Moderate / Low	Low (negative scaled value)
Total_Cards	High count	Average count	Low (1–2 cards)
Online_Visits	Highest engagement	Low	Moderate
Bank_Visits	Low	Highest engagement	Low
Calls_Made	Low	Low	Highest volume

Hierarchical Clustering

Hierarchical clustering with different linkage methods:

Single Linkage: 0.6993
Complete Linkage: 0.8306
Average Linkage: 0.8789
Centroid Linkage: 0.8742
Ward Linkage: 0.7466

Plot dendrograms for each linkage method and Check cophenetic correlation for each linkage method:



online user segment aligns with the same online users identified in the K-Means analysis.

his split effectively distinguishes users who prefer online transactions from do not.

As observed previously, online users tend to have more credit cards and higher credit limits.

It is worth mentioning that when attempting 3, 4, 5, or 6 clusters, all but two clusters contained fewer than five records, making them too small to provide meaningful insights.

K-means vs Hierarchical Clustering

Outliers

K-means: Outliers are problematic. Because the algorithm tries to minimize the distance from the centroid, an outlier can "pull" the centroid away from the actual dense center of the group.

Hierarchical: Depending on the Linkage Method, it handles outliers differently. Single Linkage is very sensitive (outliers stay separate for a long time), while Ward's Method tends to force outliers into clusters more like K-means does.

Clusters

K-means creates clusters of roughly similar density. If we have one very large group and one very tiny group, K-means might split the large group just to maintain balance.

Hierarchical allows us to see the "ancestry" of a point. We can see exactly at what distance a "Support-Seeker" point becomes similar enough to be merged with a "Traditionalist" point.

Scores :

To determine which method "won" for our data, look at these two specific scores:

Silhouette Score: If the K-means silhouette score is higher than the Hierarchical score (using the same number of clusters), it means our data naturally forms spherical, well-separated groups.

Cophenetic Correlation: Only used for Hierarchical. If this score is above 0.75, dendrogram is a high-quality representation of the actual distances between our customers.